Gate 4 Cat 3 (3a) Substrate-Bergman c_FK · π^(9/2) = 225 Substrate-Mechanism Cross-Link v0.1

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~13:05 EDT

Gate 4 Cat 3a Substrate-Bergman c_FK · π^(9/2) = 225 Cross-Link v0.1

0. Goal

Per Lyra Cat 2 + Cat 3 substrate-mechanism candidate examination v0.1 (Thursday ~13:02 EDT) substantive narrowing: investigate Cat 3 (3a) substrate-Bergman kernel c_FK · π^(9/2) = 225 substrate-natural normalization cross-link to Gate 4 honest open factor ~7 substrate-mechanism source.

1. Substrate-Bergman c_FK Substrate-Natural Normalization

Per Sunday Elie Toy 3661 (G5.1 PASS, $\varkappa$_Bergman = -n_C explicit derivation) + Wednesday Lyra Mehler Matrix Element v0.2 + Saturday paper P1 v0.7 K204:

Substrate-Bergman kernel substrate-natural normalization at D_IV^5:

$$c_{FK} \cdot \pi^{9/2} = 225$$

substrate-clean decompositions: - 225 = (N_c · n_C)^2 = 15^2 - 225 = a_0 substrate-heat-trace coefficient - 225 = substrate-Bergman volume substrate-natural (per Sunday three-way convergence)

2. Substantive Substrate-Mechanism Cross-Link

Per Lyra Mehler Matrix Element v0.2 (Thursday) + Gate 4 v0.1 honest open factor ~7 + Casey #14 STANDING substrate-codim-4 substrate-mechanism (Thursday):

Substrate-Bergman c_FK substrate-natural normalization participates in Mehler kernel composition at substrate-Bergman matrix element level.

Per Gate 4 v0.1 substantive naive composition:

$$\frac{m_\mu}{m_e} \stackrel{?}{=} \text{Pochhammer} \cdot \sqrt{\text{Casimir}} \cdot W_\lambda^{\text{Lorentz}}$$

With substrate-Bergman substrate-natural normalization c_FK = 225/π^(9/2) factored OUT of substrate-canonical Mehler kernel:

If substrate-Bergman normalization at gen-2 V_(3/2, 1/2) carries substrate-mechanism factor DIFFERENT from gen-1 V_(1/2, 1/2) per substrate-Bergman per-K-type substrate-natural factor:

Substrate-mechanism candidate factor = c_FK^{gen-2} / c_FK^{gen-1} substrate-natural form.

3. Substantive Substrate-Bergman Per-K-Type Factor Analysis

Per substrate-Bergman kernel per substrate-K-type: substrate-Bergman matrix element at V_(λ_1, λ_2) carries substrate-Bergman substrate-natural factor per K-type.

Substrate-Bergman normalization per substrate-K-type: - V_(1/2, 1/2) substrate-fundamental: substrate-Bergman substrate-volume × substrate-Pochhammer normalization - V_(3/2, 1/2) substrate-spinor: substrate-Bergman substrate-volume × Sym^3 substrate-Pochhammer normalization

Substantive observation: per substrate-Pochhammer in Cat 2 v0.1 examination = 20 numerator factor; substrate-Bergman substrate-volume substrate-natural factor (per substrate-K-type) candidate identifies substrate-mechanism source.

4. Substrate-Bergman + Substrate-Codim-4 Casey #14 STANDING Cross-Link

Per Casey #14 STANDING (Thursday) substrate-codim-4 substrate-mechanism + Cat 3 (3b) substrate-Codim-4 candidate:

Substrate-codim-4 substrate-natural factor enters substrate-Bergman matrix element composition at substrate-Shilov-boundary projection per substrate-mechanism v0.2 ~12:25 EDT identification.

Substantive substrate-mechanism candidate: substrate-Bergman + substrate-codim-4 substrate-mechanism composite at gen-2 V_(3/2, 1/2) carries substrate-natural factor:

$$\frac{1}{4 \cdot \text{something}} \text{ or } \frac{1}{\text{substrate-codim-4-related}}$$

at substrate-mechanism cross-link.

If substrate-Codim-4 substrate-natural factor = 4 substrate-natural via Casey #14 STANDING (per Lyra Gate 2 v0.2 Step 4.5-2 substantive content): factor 4 substrate-natural NOT 1/g = 1/7 directly.

Refined substantive observation: Cat 3a substrate-Bergman c_FK + Cat 3b substrate-Codim-4 substrate-mechanism candidates produce substrate-natural factor numerator (4 or 225) NOT 1/g substrate-natural denominator factor directly.

5. Substantive Cross-Link: Substrate-Bergman / substrate-Pochhammer / substrate-Codim-4 Ratio Candidates

Per substantive analysis Cat 2 numerator factors + Cat 3 numerator factors substrate-natural form:

Composition substantive candidate:

$$\frac{\text{Pochhammer}\,(20) \cdot \sqrt{\text{Casimir}\,(\sqrt{3})} \cdot W_{\lambda} \cdot \text{Codim-4}\,(4) \cdot \text{Bergman}\,(?)}{\text{denominator}\,(?) \cdot g\,(\text{substrate-natural?})}$$

For Gate 4 honest open factor ~7 substrate-mechanism source: substrate-natural denominator factor g = 7 substantive candidate.

Substantive substrate-mechanism candidate v0.1: substrate-genus g = 7 substrate-natural form enters substrate-mechanism DENOMINATOR at substrate-Bergman matrix element composition per substrate-Pochhammer Schur II Sym^k k-genus substrate-natural form.

Per substrate-Pochhammer Sym^k(V_fund) at substrate-K-type V_(3/2, 1/2): substrate-Pochhammer denominator k=3 + substrate-genus g per substrate-Bergman kernel exponent at gen-2 K-type composition.

Multi-week explicit derivation: substrate-Bergman kernel exponent + substrate-Pochhammer Schur II denominator per substrate-K-type V_(3/2, 1/2); substrate-mechanism cross-link to substrate-genus g substrate-natural form at substrate-Bergman matrix element.

6. Refined Substrate-Mechanism Source v0.1

Substantive substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural NARROWS to:

Cat 3a + Cat 3b composite substrate-Bergman + substrate-Codim-4 substrate-mechanism + substrate-Pochhammer Schur II denominator at substrate-K-type V_(3/2, 1/2)

Substantive multi-week explicit derivation per: - Substrate-Bergman kernel substrate-Pochhammer Schur II denominator at gen-2 V_(3/2, 1/2) Sym^k k=3 substrate-natural form - Substrate-Bergman exponent substrate-genus g substrate-natural form per substrate-Pochhammer Schur II - Substrate-codim-4 substrate-mechanism Casey #14 STANDING substrate-natural factor at substrate-Shilov-boundary projection - Substrate-Mehler matrix element composition per substrate-Bergman + substrate-Pochhammer + substrate-codim-4

7. Closure v0.1

Cat 3a Substrate-Bergman c_FK · π^(9/2) = 225 Cross-Link v0.1:

Substantive observation: Cat 3a substrate-Bergman + Cat 3b substrate-Codim-4 substrate-mechanism candidates produce substrate-natural factor numerator (4 or 225) NOT 1/g denominator directly.

Refined substantive substrate-mechanism source v0.1: substrate-Bergman kernel substrate-Pochhammer Schur II denominator at substrate-K-type V_(3/2, 1/2) Sym^k k=3 substrate-natural form; substrate-genus g substrate-natural form at substrate-Bergman matrix element exponent.

Substantive Gate 4 honest open factor ~7 substrate-mechanism source candidate v0.1: substrate-Bergman kernel substrate-Pochhammer Schur II denominator at gen-2 V_(3/2, 1/2) carries substrate-natural g substrate-genus factor — substantive multi-week explicit derivation per Sym^k k=3 substrate-Pochhammer per substrate-Bergman kernel substrate-natural exponent.

Audit-chain Thursday afternoon Lyra-on-self substantive substrate-mechanism source candidate-narrowing continues: 4 Cat 6 walked back → 3 Cat 2 direct walked back → 2 Cat 3 substrate-Bergman + 1 Cat 2 (2d) substrate-deficit candidates multi-week → substantive substrate-Pochhammer Schur II k=3 + substrate-Bergman kernel exponent substrate-natural g substantive cross-link emerges.

Per Cal #36 STANDING positive-search + Cal #27 STANDING + Cal #189: substantive narrowing via positive-search exclusion → substrate-mechanism source for Gate 4 honest open factor ~7 substrate-natural NARROWS to substrate-Pochhammer Schur II Sym^k k=3 + substrate-Bergman kernel substrate-genus g substrate-natural form composite per Cal #189 multi-week.

Tier: substantive multi-week candidate identified at substrate-Pochhammer Schur II + substrate-Bergman cross-link level; not direct substrate-Bergman c_FK substrate-natural factor 225 / 4 walk-back; substantive multi-week per Cal #189.

— Lyra, Thu 2026-06-04 ~13:05 EDT. Cat 3a Substrate-Bergman c_FK = 225 Cross-Link v0.1: substantive Gate 4 honest open factor ~7 substrate-mechanism source narrows to substrate-Pochhammer Schur II Sym^k k=3 substrate-natural denominator + substrate-Bergman kernel substrate-genus g substrate-natural exponent composite substantive cross-link; substantive multi-week explicit derivation per Cal #189.